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Multi-scale enhanced graph convolutional network for mild cognitive impairment detection

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ABSTRACT

As an early stage of Alzheimer's disease (AD), mild cognitive impairment (MCI) is able to be detected by analyzing the brain connectivity networks. For this reason, we devise a new framework via multi-scale enhanced graph convolutional network (MSE-GCN) for MCI detection, which integrates the structural and functional information from the diffusion tensor imaging (DTI) and resting-state functional magnetic resonance imaging (R-fMRI), respectively. Specifically, both information in the brain connective networks is first integrated based on the local weighted clustering coefficients (LWCC), which is concatenated as the feature vector for representing a population graph's vertice. Simultaneously, the gender and age information in each subject are integrated with the structural and functional features to construct a sparse graph. Then, various parallel graph convolutional network (GCN) layers with multiple inputs are designed from the embedding from random walk embeddings in the GCN to identify the essential MCI graph information. Finally, all GCN layers' outputs are concatenated via the fully connection layer to perform disease detection. The experimental results on the public Alzheimer's Disease Neuroimaging Initiative (ADNI) database show that our method is promising to detect MCI and superior to other competing algorithms, with a mean classification accuracy of 90.39% in the detection tasks.

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1. Introduction

As an early stage of Alzheimer's disease (AD), mild cognitive impairment (MCI) has a 10%–15% conversion rate each year in 5 years [1]. Early detection is the key to slow down the progression of AD. Therefore, learning an effective prediction model for MCI detection receives the increasing attention [2–5]. Early detection of MCI can be achieved by identifying changes in imaging markers. As a new MRI technology, the resting-state functional MRI (R-fMRI) [6] has been commonly applied for brain disease studies. R-fMRI can check functional integration and separation of brain networks, which provides effective functional connectivity (FC) information for detecting early MCI (EMCI) [7]. Functional connectivity (FC) can represent the temporal correlation of blood-oxygenation-level-dependent (BOLD) time series between two brain regions and FC approaches have been successfully used to detect MCI [8–10]. Also, by revealing the variations in anisotropy values, connection number and length of fiber bundle, diffusion tensor imaging (DTI) [11] is able to reveal the structural information from fiber bundles of the white matter via the detection of the brain's opposite values for EMCI detection.

Most of the previous research mainly focused on the FC constructed from R-fMRI time series among different brain regions [12, 13] by calculating the co-activation level. For example, Schwab et al. [14] developed a dynamic graph model for the construction of brain networks and obtained better results than static brain networks. Wee et al. [15] constructed brain functional connectivity network through sparse learning method via group constraint, which achieved quite good accuracy in MCI classification. Meanwhile, some studies also used DTI to construct structural brain networks, which used graph theory to analyze the nature of the networks and explain the relationship between changes in brain con-

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nections and brain diseases [16]. In the course of studying the general relationship between functional and structural connectivity, it is discovered that the global brain area analysis shows a significant overall consistency between the two types of connectivity. However, the analysis in the region of interests (ROIs) shows that the connectivity between regions breaks such consistency [17]. This indicates that functional and structural networks are complementary. Therefore, to overcome the limitation of subtle differences in neuroimaging features, it is increasingly popular to use multi-modal data to describe or strengthen the features from multiple sources [18–22]. For example, Li et al. [23] proposed a super group-constraint orthogonal positive regression to consider the constraint relationship between DTI and R-fMRI to construct a brain network, and then used the elastic multilayer perceptron classifier for MCI recognition. Liu et al. [24] proposed a multi-perspective model to learn multi-modal data characteristics for MCI classification. However, the high dimension of FC and structural connectivity (SC) make the deep learning model suffer from overfitting and result in poor performance [25]. To address this problem, the feature selection is an effective way. For example, Song et al. [26]. extracted the upper triangular matrix elements from FC and SC, and then used the recursive feature elimination (RFE) [27] to select features. Zhu et al. reduced the feature dimension by the method of Lasso [28]. However, these methods contain too many parameters, which make them difficult to tune for clinical application. To overcome these problems, local weighted clustering coefficients (LWCC) can measure the degree of node clustering and can quantify the degree to which the neighbor nodes gather from a network, and this method does not need many parameters [29]. Inspired by these methods, in this study, we extract both FC and SC from one subject and concatenate them via LWCC to perform multi-modal fusion for better feature representation.

Recently, some works report that a higher incidence of MCI occurs in females than in males and MCI risk increases with age [30]. Furthermore, graph learning is able to utilize these information in MCI detection task effectively [31]. Therefore, combination of demographic information (e.g., age and gender) can boost the EMCI detection performance [32, 33]. Inspired by this, all subjects are represented by using a sparse graph (denoted as an MCI-graph), in which each vertex denotes a subject's features and the edges contain both subject's features and demographic information (e.g., gender and age). How to effectively learn an MCI-graph and predict the node of MCI-graph label are challenging tasks. Graph kernels have been successfully applied in many graph classification tasks in the graph learning field [8]. These methods are able to encode graphs or nodes into feature vector spaces in a mapping way. However, graph kernel methods have computational bottlenecks because of the pair-wise similarity calculation [34]. By contrast, as an end-to-end training model, graph convolutional network (GCN) offers a fruitful and progressive way, which can directly perform EMCI detection by graph representations [35]. It is worth noting that suboptimal detection performance may be obtained by the existing GCN methods for its limited receptive field. For this reason, lots of researchers attempted to improve GCN's receptive field for effective detection performance [36]. For example, Xu et al. [37] have found that stacking more GCN layers can increase the receptive field, but it will result in over-smoothing and degrade the performance. Recently, graph embedding via unsupervised learning has benefits of learning the information by the random walks due to the promising performance [38]. As a result, Sami et al. [39] developed a network of GCNs to integrate two lines of work (i.e., random walks and semi-supervision) into their model. Yang et al. [40]. explored the effectives of random walks used in the graph. Giannis et al. [41]. proposed a network called random walk graph neural work and found that using random kernel in graph representation can get better performance. Also, it is known that the multi-scale GCN (MS-GCN) not only can boost the network's receptive field, but also effectively take advantage of the high-order information of the brain with random walk. Inspired by this, we develop a multi-scale enhanced GCN (MSE-GCN) for booting the traditional GCNs' receptive field, which detects MCI using random walk algorithm.

In this study, an MSE-GCN is designed to exploit each subject's differences and information associated with subjects for MCI detection. First, we build a FC network from the R-fMRI and a SC network from DTI. Then, both FC and SC information are utilized to enhance single model's performance via complementary data discovery and fusion. To expand the receptive field of GCN and learn the higher order representation of features, various parallel GCN layers are designed with different inputs, which can analyze MCI graph effectively by random walk embedding theory. We validate the effectiveness of our proposed model using the publicly available Alzheimer's Disease Neuroimaging Initiative (ADNI) database. The proposed model obtains quite remarkable performance for MCI detection and is superior to the related competing methods. The difference between our previous work [42] and this work is four-fold: 1) We add a three classification experiments to verify the effectiveness of our model. 2) We do experiment to verify the effectiveness of random walk, and find the best step of random walk. 3) We add experiment to verify the performance of feature selection LWCC is better than other feature selection in our four works of classification. 4) We add more visual charts to validate our model. Overall, our main contributions are three-fold:

1) We design an MCI-graph framework which integrates both non-image information (i.e., subject's age and gender) and image information (i.e., FC and SC features).
2) We devise various parallel GCN layers using multiple inputs from the random walk embedding theory, which is able to identify the essential MCI information from the GCN graph embedding.
3) We fuse the information from the functional networks and structural networks using an MSE-GCN model to improve prediction performance.

2. Methodology

In this study, a new framework is devised for MCI detection. As shown in Fig. 1, the proposed method contains three modules: (1) Brain connectivity network construction, (2) Multi-modal fusion and population graph construction, and (3) Multi-scale GCN for MCI detection. To be specific, a non-overlapping sliding window is used to split the R-fMRI data into different subseries, which can get dynamic FCs by computing the Pearson correlation (PC) coefficient from the paired subseries. Moreover, a structural network is established from each pair of brain regions' connection of the white matter fiber bundles. Both LWCC of SC and FC are then computed and combined into a feature vector for the representation of a phenotypic graph's vertices, where graph edges denote the subjects' similarity. The demographic information is incorporated as well. Lastly, we adopt the MSE-GCN algorithm for MCI detection. Table 1 summarizes the important notations.

2.1. LWCC extraction and data fusion

Extracting discriminative features and multimodal fusion are key steps in the MCI detection. In most prior works, the upper half matrix of FC/SC from a subject is usually used as the feature vector [43]. However, this not only leads to destroy the structure of the original network, but also brings high dimensional features, which is not beneficial for the MCI detection. To solve it, we extract the LWCC $\boldsymbol{F} = \{f_i | (i = 1, 2, \ldots, r)\}$ from a brain connectivity

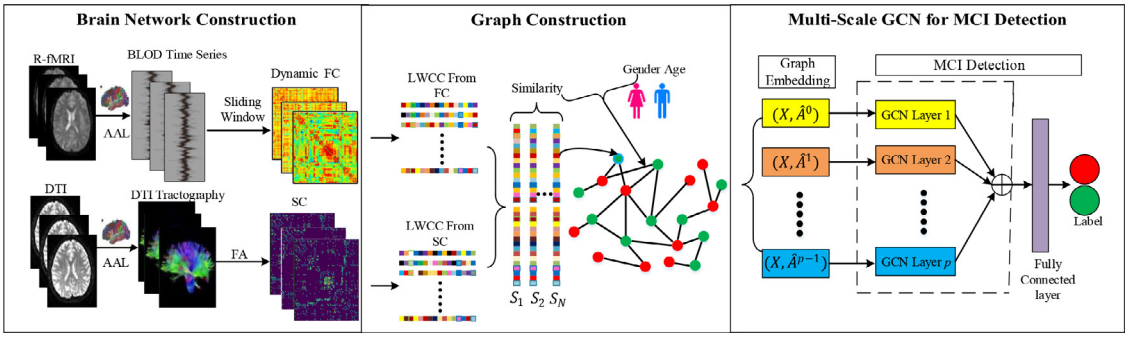

Fig. 1. Illustration of the overall architecture. We first construct the FC network from R-fMRI and SC network from DTI. Then we extract features by LWCC from the FC and SC separately and fuse the features to construct a multi-modal graph. At last, the multi-modal graph is fed to our Multi-Scale GCN for MCI detection.

Table 1
Commonly used notations in this paper.

Notation	Description
$\odot$	Element-wise product
K	Chebyshev polynomial of different orders
$\boldsymbol{X} \in \boldsymbol{R}^{N\times d}$	The feature matrix of a graph
$\boldsymbol{A} \in \boldsymbol{R}^{N\times N}$	The graph adjacency matrix
$\boldsymbol{G}$	A graph
$\boldsymbol{D} \in \boldsymbol{R}^{N\times N}$	The degree matrix of $\boldsymbol{A}$
N	The number of nodes
$d = (\lambda + 1)r$	The dimension of a node feature vector
$\sigma(.)$	The activation function
$\|\|.\|\|$	ℓ_2-norm of vector x
E	The set of edges in a graph
$E\|Y\|$	The mathematical expectation of Y
$\mathcal{L}$	Row normalization matrix
$\mathcal{P} \in \boldsymbol{R}^{N\times N}$	Non-image distance metric
$\boldsymbol{S_n} \in \boldsymbol{R}^{d}$	Feature vector of one subject
$\boldsymbol{S} \in \boldsymbol{R}^{N\times N}$	Similarity matrix

matrix $\boldsymbol{W} \in \mathbb{R}^{r\times r}$ (e.g., FCs or SCs) to obtain the corresponding feature vectors. Specifically, the weights of the edge connecting vertex i and vertex j for each network (i.e., the i th node) are denoted as

$$f_i = \frac{2\sum_{j:j\in\varepsilon_i}\left(w_{ij}\right)^{\frac{1}{3}}}{|\varepsilon_i|(|\varepsilon_i|-1)}, \tag{1}$$

where ε_i denotes the set of vertices directly connected to vertex i, $|\varepsilon_i|$ is the number of elements in ε_i, and w_{ij} is the weight of FC or SC. For example, we can obtain $\lambda + 1$ feature vectors F composed of LWCC when λ FCs and one SC are obtained from a subject. Then, we concatenate $\lambda + 1$ feature vectors F to get one feature vector: $S_N = \{F_1, F_2, \ldots, F_{\lambda+1}\}$, where N is the subject's ID. The proposed method not only can decrease the dimension of the features, but also can quantify each node's clustering in the weighted brain network.

2.2. Graph construction

It is important to define an effective graph to detect MCI. Hence, both image data (e.g., R-fMRI and DTI) and non-image data (e.g., gender and age) are utilized to construct a sparse graph to represent our data's intrinsic structure. In this study, integration of both information can discover the similarity among features of every subject and assist the label prediction. Let $G(V, E, \boldsymbol{A})$ denote the proposed undirected graph, where V means the extracted feature vector of the subjects, edge E represents the similarity of the subjects and encodes the non-image information, $\boldsymbol{A}$ is the adjacency matrix, which means the vertex connectivity calculated by the sample point distance metric using similarity matrix $\boldsymbol{S}$. For subjects n and m $(n, m = 1, 2, \ldots, N)$, $\boldsymbol{A}$ is defined as

$$\boldsymbol{A_{nm}} = \boldsymbol{S}(S_n, S_m)\sum_{g=1}^{g}\mathcal{P}(M_g(S_n), M_g(S_m)), \tag{2}$$

where g denotes non-image metric $M = \{M_g\}$, and $\boldsymbol{S}(S_n, S_m)$ denotes the subject's feature similarity. The distance similarity metrics are used to calculate the distance of subject n and m. For distance between vector S_n and S_m, it is denoted as

$$\boldsymbol{S}(v_n, v_m) = \frac{(S_n - \bar{S}_n)\odot(S_m - \bar{S}_m)}{\left\|\left(S_n - \bar{S}_n\right)\right\|_2\left\|(S_m - \bar{S}_m)_2\right\|} \tag{3}$$

where $\bar{S}_n$is the average of S_n and $\odot$ means the dot product. $\mathcal{P}()$ denotes the distance between n and m, which is denoted as

$$\mathcal{P}(M_g(s_n), M_g(s_m)) = \{^{1,|M_{age}\ (s_n)-M_{age}\ (s_m)|<\beta\ \&\ M_{gender}\ (s_n)=M_{gender}\ (s_m)}_{0,otherwise}, \tag{4}$$

where β is the standard deviation of the subjects' ages.

2.3. Spectral convolution

In the spectral graph theory, the properties of a graph are studies from the eigen-values and eigen-vectors of its Laplacian matrix. It was first proposed by graph signal processing researchers to define Fourier transformation on the graph [44]. Then it is used to define the convolution on the graph. The spectral approach offers a good graph localization operator using the Kronecker product applied in the spectral domain [45]. The graph Laplacian matrix after normalization is denoted as

$$\boldsymbol{L} = \boldsymbol{I_N} - \boldsymbol{D}^{-\frac{1}{2}}\boldsymbol{A}\boldsymbol{D}^{-\frac{1}{2}} = \boldsymbol{D}^{-\frac{1}{2}}(\boldsymbol{D} - \boldsymbol{A})\boldsymbol{D}^{-\frac{1}{2}} = \boldsymbol{U}\boldsymbol{\Lambda}\boldsymbol{U}^T, \tag{5}$$

where $\boldsymbol{U}$ is an eigen-matrix composed of Fourier basis and $\boldsymbol{\Lambda}$ is a diagonal matrix of its eigen-values, $\boldsymbol{D}$ $(d_n = \sum_{m=1}^{N} a_{nm} \in \boldsymbol{D})$ is denoted as the diagonal degree matrix denoting the vertex connections' number. Therefore, the spectral graph convolution operator is denoted as $g_\mu = diag(\mu)$ and

$$g_\mu * \boldsymbol{X} = g_\mu(\boldsymbol{L})\boldsymbol{X} = \boldsymbol{U}g_\mu(\boldsymbol{\Lambda})\boldsymbol{U}^T\boldsymbol{X}, \tag{6}$$

where the parameter μ is a vector of Fourier coefficients in the Fourier domain and the graph Fourier transform of a signal $\boldsymbol{X}$ is defined as $\boldsymbol{U}^T\boldsymbol{X}$. $g_\mu(\boldsymbol{\Lambda})$ is a non-parametric filter, which can be expanded in terms of polynomials to its K-th order, i.e.,

$$g_\mu(\boldsymbol{\Lambda}) = \sum_{k=0}^{K-1}\mu_k\boldsymbol{\Lambda}^k, \tag{7}$$

where the parameter $\mu \in \mathbb{R}^K$ is a vector of Chebyshev polynomial coefficients. The value at node i of the filter g_μ centered at node

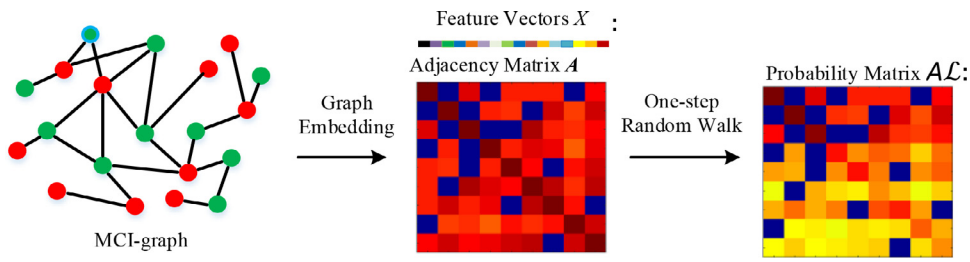

Fig. 2. The diagram of one-step random walk in a graph. In the process of one-step random walk, we designate a node on the graph as the starting node. Then, the nodes around it should be randomly selected to move, and the current vertex should be taken as the starting point to move to the next vertex random. The process should be repeated continuously to get a probability matrix.

j is given by $(g_\mu(\boldsymbol{L})\delta_j)_i = (g_\mu(\boldsymbol{L}))_{i,j} = \sum_k \mu_k(\boldsymbol{L}^k)_{i,j}$, where the kernel is localized via a convolution with a Kronecker delta function $\delta_j \in \mathbb{R}^n$. In fact, this filter is well-approximated to the truncation expansion of the Chebyshev polynomial, which is denoted as

$$g_\mu(\boldsymbol{\Lambda}) = \sum_{k=0}^{K-1} \mu_k T_k(\tilde{\boldsymbol{\Lambda}}), \tag{8}$$

where $\mu \in \mathbb{R}^K$ is a parameter of Chebyshev coefficients' vector, $T_k(\tilde{\boldsymbol{\Lambda}})$ denotes the polynomial of order K in Chebyshev coefficient, and $\tilde{\boldsymbol{\Lambda}} = 2\boldsymbol{\Lambda}/\lambda_{max} - \boldsymbol{I}_n$.

The feature map is implemented to calculate the spatial feature from the weighted sum of the central pixel and the adjacent pixel. With the normalized adjacency matrix $\hat{\boldsymbol{A}} = \boldsymbol{D}^{-\frac{1}{2}}\boldsymbol{A}\boldsymbol{D}^{-\frac{1}{2}}$, the network layer is denoted as

$$\boldsymbol{H}^{(l+1)} = f\left(\boldsymbol{H}^{(l)},\ \hat{\boldsymbol{A}}\right), \tag{9}$$

$$f\left(\boldsymbol{H}^{(l)}, \hat{\boldsymbol{A}}\right) = \sigma\left(\hat{\boldsymbol{A}}\boldsymbol{H}^{(l)}\boldsymbol{W}^{(l)}\right), \tag{10}$$

where $\boldsymbol{W}^{(l)}$ is the weight matrix of the l-th layer neural networks, $\boldsymbol{H}^{(l)}$ is the input of l-th layer, and $\sigma(\cdot)$ is the non-linear activation function. Therefore, the two-layer GCN model is calculated as

$$Z = f\left(\boldsymbol{X}; \hat{\boldsymbol{A}}\right) = \sigma_1\left(\hat{\boldsymbol{A}}\sigma_0\left(\hat{\boldsymbol{A}}\boldsymbol{X}\boldsymbol{W}^{(0)}\right)\boldsymbol{W}^{(1)}\right). \tag{11}$$

2.4. Multi-scale GCN

Recent study shows that traditional GCNs may lead to suboptimal performance of MCI detection due to the limitation of receptive field. Also, unsupervised learning of graph embedding has benefited from the information contained in random walks. Therefore, in this study, we use random walk embedding technique on graphs to obtain node representations. Fig. 2 shows the diagram of one-step random walk in a graph.

Random walk [46] is a special form of Markov chain, which represents the probability of the information from one vertex going to another vertex in a graph. This probability matrix is statistically known as a co-occurrence matrix $\boldsymbol{Y}$. Therefore, random walk essentially transforms the vertices of a graph into a continuous vector space. The expectation on $\boldsymbol{Y}$ is defined as

$$\mathcal{E}|\boldsymbol{Y}| = \mathcal{E}_{p\sim Q}\left[\left(\boldsymbol{D}^{-1}\boldsymbol{A}\right)^q\right] = \mathcal{E}_{p\sim Q}\left[(\mathcal{L})^p\right], \tag{12}$$

where p is number of steps of the random walk, $\mathcal{L}$ is a row normalized transition matrix. Besides, Q abides by super-parameters of the random walk. Doing one-step random walk on the graph is shown in Fig. 2, where $\mathcal{L}$ is a row normalization transition matrix.

In a particular case, the GCN can simulate the principle of a random walk. For example, supposing the activation function $\sigma_0(z) = z$ is identical and $\boldsymbol{W}^{(0)}$ is an identity matrix, then Eq. (11) can be reformulated as

$$Z = f\left(\boldsymbol{X}; \hat{\boldsymbol{A}}\right) = \sigma_1\left(\hat{\boldsymbol{A}}\hat{\boldsymbol{A}}\boldsymbol{X}\boldsymbol{W}^{(1)}\right) = \sigma_1\left(\hat{\boldsymbol{A}}^2\boldsymbol{X}\boldsymbol{W}^{(1)}\right). \tag{13}$$

Table 2
Summary of the data used in this study.

Group	EMCI (77)	LMCI (40)	NC (67)
Male/Female	39 M/38F	25 M/15F	28 M/39F
Age (mean±SD)	74.8±6.1	74.5±6.9	75.4±6.5

As can be seen from Eq. (13), Z can be regarded as a GCN layer with input $(X, \hat{\boldsymbol{A}}^2)$, where $\hat{\boldsymbol{A}}^2$ is denoted as:

$$\begin{aligned}\hat{\boldsymbol{A}}^2 &= \left(\boldsymbol{D}^{-\frac{1}{2}}\boldsymbol{A}\boldsymbol{D}^{-\frac{1}{2}}\right)\left(\boldsymbol{D}^{-\frac{1}{2}}\boldsymbol{A}\boldsymbol{D}^{-\frac{1}{2}}\right) \\ &= \boldsymbol{D}^{-\frac{1}{2}}\boldsymbol{A}(\boldsymbol{D}^{-1}\boldsymbol{A})\boldsymbol{D}^{-\frac{1}{2}} = \boldsymbol{D}^{-\frac{1}{2}}\boldsymbol{A}\mathcal{L}\boldsymbol{D}^{-\frac{1}{2}}.\end{aligned} \tag{14}$$

In this way, the GCN model can effectively perform one-step random walk. Next, we expand the above formula to p-degree polynomial as

$$\hat{\boldsymbol{A}}^p = \boldsymbol{D}^{-\frac{1}{2}}\boldsymbol{A}\mathcal{L}^{p-1}\boldsymbol{D}^{-\frac{1}{2}} \tag{15}$$

where p-step random walk will allow every node to accumulate information from its neighbors within distance p. The p-th power contains statistics from the p-th step of a random walk on the graph. Therefore, our MSE-GCN model is able to combine information from multiple graph scales. Meanwhile, using p power of the normalized adjacency matrix $\hat{\boldsymbol{A}}$ as input can explore the higher order similarity of features, which is beneficial to understand diseases. Motivated by this, we further design p parallel GCN layers. The input of p-th GCN layer is $(X, \hat{A}^{p-1})$ and the output of each GCN layer is a matrix $\mathbb{R}^{N\times C_p}$, where C_p is the latent dimensionality. Finally, we concatenate the output of all the GCN layers into the full connection layer to perform MCI detection.

3. Materials

3.1. Data acquisition

We use a total number of 184 subjects in this study. 40 late mild cognitive impairment (LMCI) patients, 77 early mild cognitive impairment (EMCI) patients, and 67 normal controls (NC) are obtained from the ADNI dataset (http://adni.loni.ucla.edu). Table 2 summarizes the subjects' information in this study.

3.2. Data preprocessing

The R-fMRI data preprocessing procedure is performed using GRETNA [47]. To keep magnetization equilibrium, we discard the first 20 acquired R-fMRI volumes of each subject. Afterward, the head-motion artifacts in the R-fMRI time-series are removed by using the first volume of each subject as a reference to register all the subsequent volumes. After registration, we use DARTEL [48] for spatial normalization and spatially smooth the dataset with a 4 mm full width half maximum Gaussian kernel followed

by band-pass filtering of frequency interval (0.01 Hz$\leq f \leq$ 0.08 Hz) to minimize the effects of low-frequency drift and high-frequency noise. Each regional mean time series is regressed against the average cerebrospinal fluid and white-matter signals as well as the six parameters from motion correction. A parcellation of the brain space into 90 region of interests is then performed by warping the automated anatomical labeling (AAL) [49] atlas to the R-fMRI images. Finally, we obtain the time series of each individual ROI of each subject via the time series over all voxels in particular ROI.

The DTI data preprocessing procedure is performed using PANDA with five steps [50]. The first step converts the DICOM format into the NIFTI format, which generates three files, i.e., bvals text files, bvecs text files, and 4D data file. These three files represent the *b* value size, *b* magnetic field gradient, and DTI of 4D data, respectively. The second step resamples the skull stripping data since DTI has a high resolution of $0.9 \times 0.9 \times 1$ mm^3. The fiber bundle tracking takes too much time. Therefore, a resampling to $2 \times 2 \times 2$ mm^3 is used to improve pretreatment efficiency. In this work, skull stripping is carried out with a parameter of 0.25 mm to extract the brain volume and reduce the influence of skull on model learning. The third step cuts the images with a parameter of 3 mm to reduce the influence of peripheral noise. The fourth step corrects head movements and eddy currents, which can reduce the influence of noises. The fifth step defines the tracking conditions and network nodes, generates the structural network matrix based on the deterministic fiber tracking, and sets the tracking stopping conditions: 1) if the crossing angle of two consecutive moving directions exceeds 45°, the tracking stops, or 2) if the fractional anisotropy (FA) value is not between 0.2 and 1, the tracking stops. Node definition requires FA images and magnetic resonance imaging (MRI) images. An AAL template is used as the segmentation template and Montreal Neurological Institute (MNI) is selected as the standard template for MRI. The FA image is registered to MRI, which is registered again to standard MNI template. The generated mapping matrix maps the AAL template to the brain of the corresponding subject by means of inverse mapping. After completing the definition of the network node, we carry out the deterministic fiber tracking to finally obtain the structural network matrix.

4. Experiments

4.1. Experiment setup

Since ADNI is a relatively small dataset, in this study, we use a leave one-out (LOO) cross-validation strategy for all experiments to conduct fair comparisons. The performance is evaluated and compared by mean values of classification accuracy (*ACC*), sensitivity (*SEN*), specificity (*SPEC*), and *F-score* (F1). For all GCN-based competing methods, we use the same model parameters: 20% dropout rate, 10^{-5}wt of the l_2 regularization term, 0.002 learning rate, and 500 features. Networks are all trained for 200 epochs. The parameters of support vector machine (SVM) [51] are: the kernel is 'sigmoid', the kernel coefficient is 0.1, the regularization parameter is 0.1, and the maximum number of iterations is 200. We set GCN with K={1,2,3,4,5} order Chebyshev polynomials. Also, grid search is applied to select optimal parameters for all methods.

4.2. Classification performance of our method

In this study, four kinds of MCI detection tasks, i.e., (1) EMCI vs. NC; (2) LMCI vs. NC; (3) LMCI vs. EMCI; and a three- category classification task (4) LMCI vs. EMCI vs.NC, are performed. We also compare our proposed method with the following methods: (1) SVM [51]; (2) Two-layer GCNs; (3) MS-GCN with the same normalized adjacency matrix [52]. Table 3 shows that the classification performance of our proposed method with other competing methods while R-fMRI, DTI and multimodal fusion data are used, respectively. For convenient viewing, we use Ours (R), Ours (D), and Ours (F) to represent the proposed method with the R-fMRI, DTI and fused data as input, respectively. Experiment result shows that GCN has the advantages in analyzing MCI compared with traditional machine learning methods. We believe that GCN can effectively use non image information in MCI detection tasks. In addition, experiments also show that increasing the receptive field of GCN is beneficial for MCI detection. Moreover, the experiments also demonstrate that the MCI detection is significantly improved with random walk graph embedding. Our analysis suggests that a random walk graph embedding may be an efficient way to extract the brain's high-order information. Fig. 3 shows the t-SNE visualization of fully-connected hidden layer of MSE-GCN when trained over MCI-graph. The ROC of different models and tasks are given in Fig. 4. In general, our method achieves better classification performance. In the four classification tasks, our proposed method has the best mean accuracy of 85.42%, 93.46%, and 92.31%, 83.11% respectively. It is found that the multi-modal fusion method achieves better classification results than single modality in the classification tasks, among which the performance of our model is better than the other competing methods.

5. Discussions

5.1. The influence of phenotypic information

An advantage of GCN algorithm is combining image information and non-image phenotypic information to predict disease. Gender and age are quite important for MCI prediction. Considering the two pieces of phenotypic information and a similarity function, we investigate the effect of them on the overall prediction performance with different combinations of phenotypic information and similarities, and the results in our four prediction tasks are shown in Fig. 5.

From Fig. 5, we observe the obvious variation of accuracies using different phenotypic information. Specifically, the performance of graph based on similarity only is the worst, whereas the performance based on similarity and both two pieces of phenotypic information (sex and age) is the best. These findings are consistent with previous studies [30].

5.2. The influence of polynomial order k

We test the effect of polynomial order K in our MSE-GCN on performance. Here, we test K={1,2,3,4,5}, and the mean classification accuracy and AUC are shown in Fig. 6. We can see that ACC value and AUC value have a small range of change, which indicates that our model is not sensitive to the order of Chebyshev polynomial K. Specifically, when $K = 4$, the model obtains the best classification results. When the order K changes from 1 to 4, the average ACC and AUC of the four detection tasks increase slightly. When the order number is larger than 4, the average ACC and AUC of the four detection tasks drop slightly. Compared with previous study, our MSE-GCN model has better stability than traditional GCN model.

5.3. The influence of random walk step p

A p-step random walk or a p-layer GCN allows every node to accumulate information from its neighbors within distance p. The former averages node feature vectors according to the random walk co-visit statistics, whereas the latter creates non-linearity and matrix multiplies at every step. We demonstrate that our models are able to use information from farther nodes. We test our

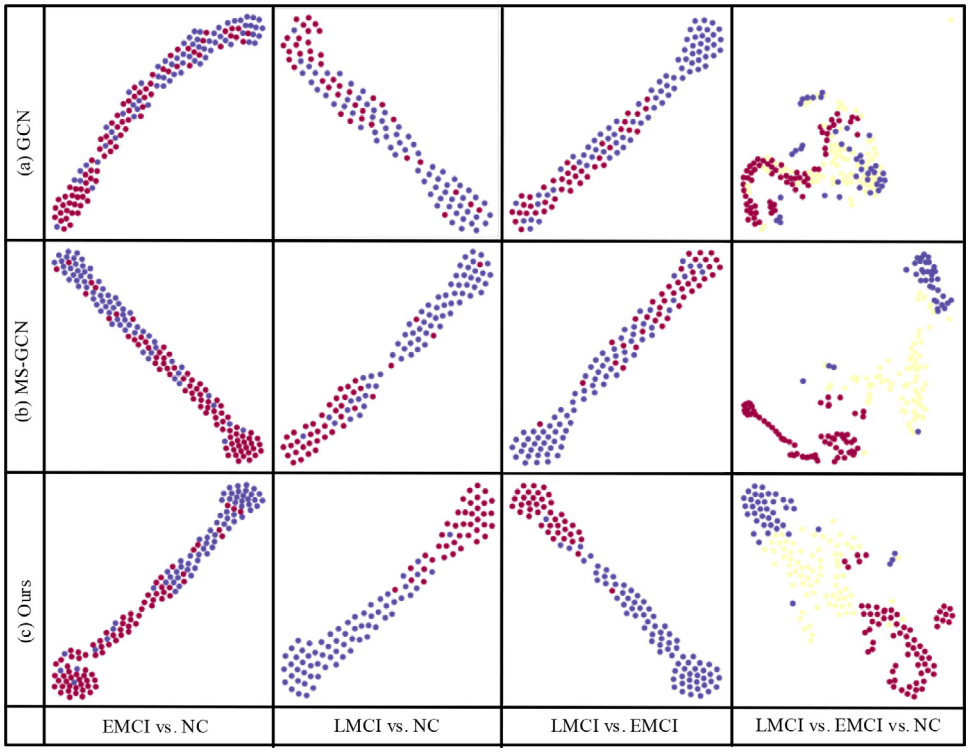

Fig. 3. The t-SNE visualization of fully-connected hidden layer of MSE-GCN when trained over MCI-graph.

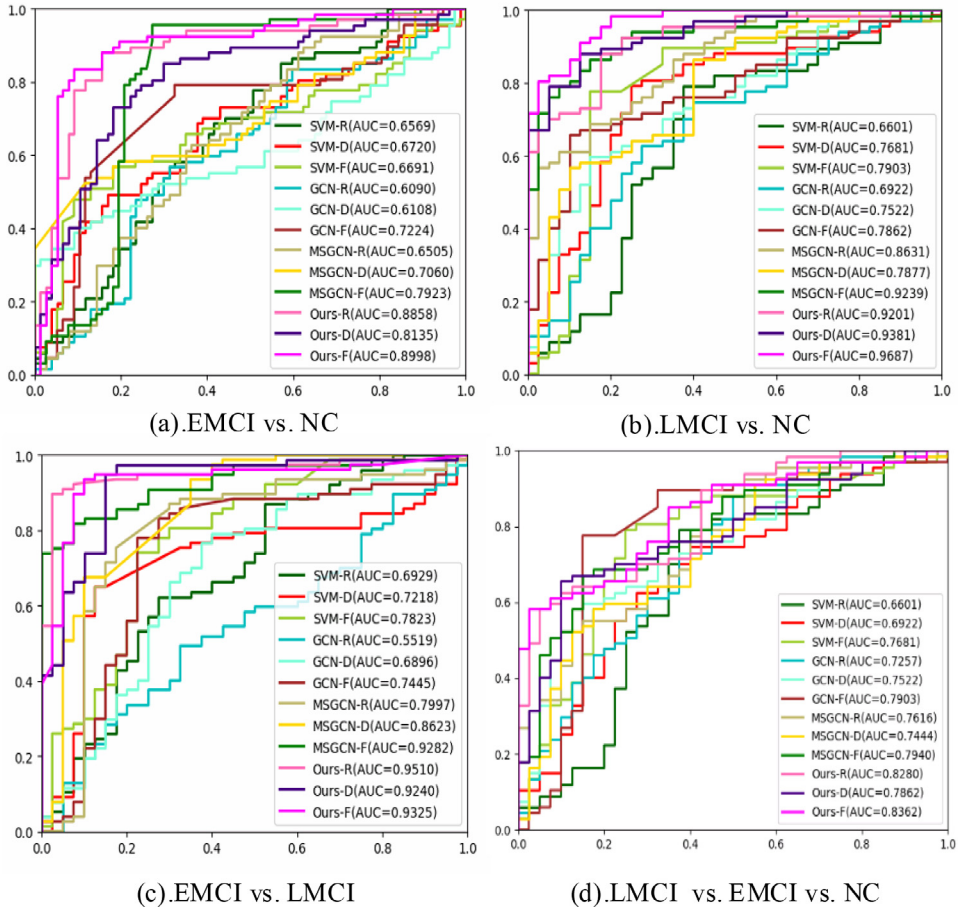

Fig. 4. The ROC of different models in each task.

Table 3
Mean detection performance of different methods in the three classification tasks (%).

Method	EMCI vs.NC				LMCI vs.NC				LMCI vs. EMCI				LMCI vs. EMCI vs. NC			
	Acc	Sen	Spec	F1	Acc	Sen	Spec	F1	Acc	Sen	Spec	F1	ACC	Sen	Spec	F1
SVM(R)	63.19	67.53	58.21	66.24	69.16	62.50	73.13	60.24	68.38	47.50	79.22	50.69	68.82	65.35	69.48	66.01
SVM(D)	63.19	75.32	49.25	68.43	75.70	72.50	77.61	69.05	72.65	67.50	75.32	62.79	70.58	68.56	72.21	69.22
SVM(F)	68.75	61.19	75.32	64.57	78.50	67.50	85.07	70.13	76.07	72.50	77.92	67.44	73.52	71.58	74.15	76.81
GCN(R)	60.42	77.92	40.30	67.80	70.09	83.58	47.50	77.78	57.26	66.23	40.00	67.11	69.41	70.89	69.21	72.57
GCN(D)	68.50	59.35	77.61	58.46	71.03	85.07	47.05	78.62	64.10	76.92	56.41	59.82	71.17	75.45	73.88	75.22
GCN(F)	71.53	67.53	76.12	71.72	73.83	86.57	52.50	80.56	75.21	84.42	57.50	81.76	74.70	76.32	72.85	79.03
MS-GCN(R)	75.69	76.12	75.32	74.45	78.63	77.50	79.22	71.25	80.34	85.71	70.00	85.16	73.52	72.86	74.58	76.16
MS-GCN(D)	70.14	65.67	74.03	67.18	74.77	74.63	75.00	78.74	79.49	87.01	65.00	84.81	72.35	70.37	74.28	74.44
MS-GCN(F)	81.25	72.73	91.04	80.58	85.05	89.55	77.50	88.24	84.62	90.91	72.50	88.61	77.64	76.25	77.96	79.40
Ours(R)	77.08	85.71	67.16	80.00	86.92	89.55	82.50	89.55	89.74	93.51	82.50	92.31	80.58	81.65	79.77	82.80
Ours(D)	81.94	89.61	73.13	84.15	89.72	92.54	85.00	91.85	90.60	94.81	82.50	92.99	79.41	80.28	79.54	78.62
Ours(F)	85.42	86.57	84.42	84.67	93.46	94.03	92.50	94.74	92.31	93.51	90.00	94.12	83.11	81.30	85.65	83.62

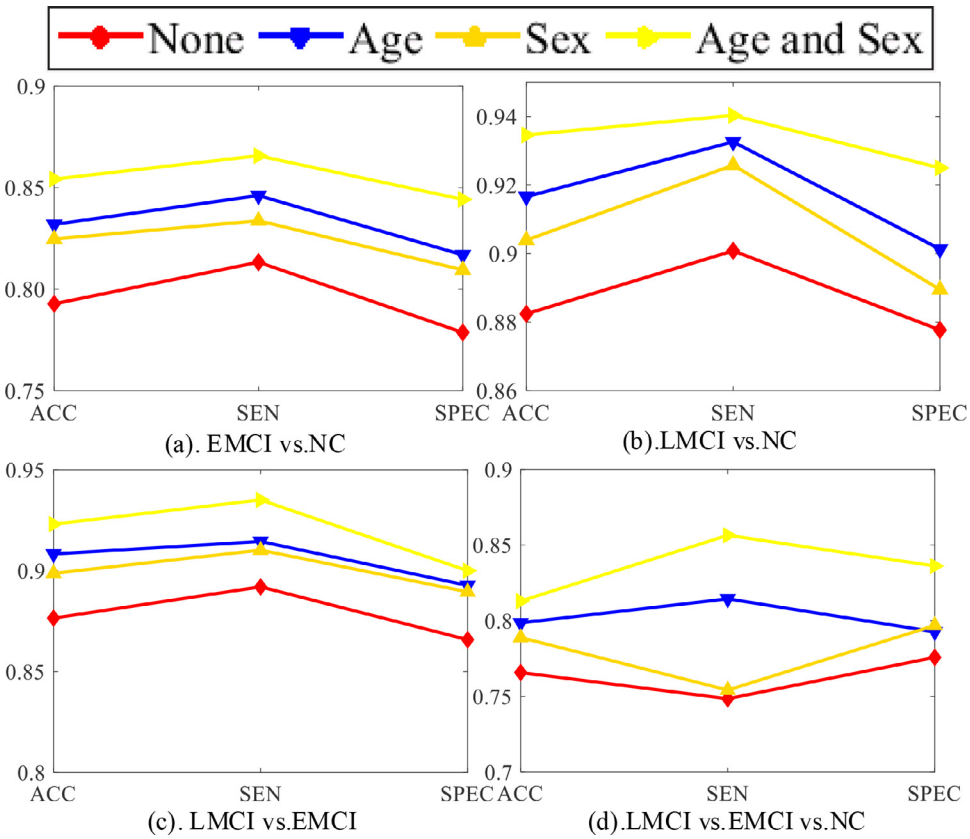

Fig. 5. Detection performance comparison of using different phenotypic information in each task.

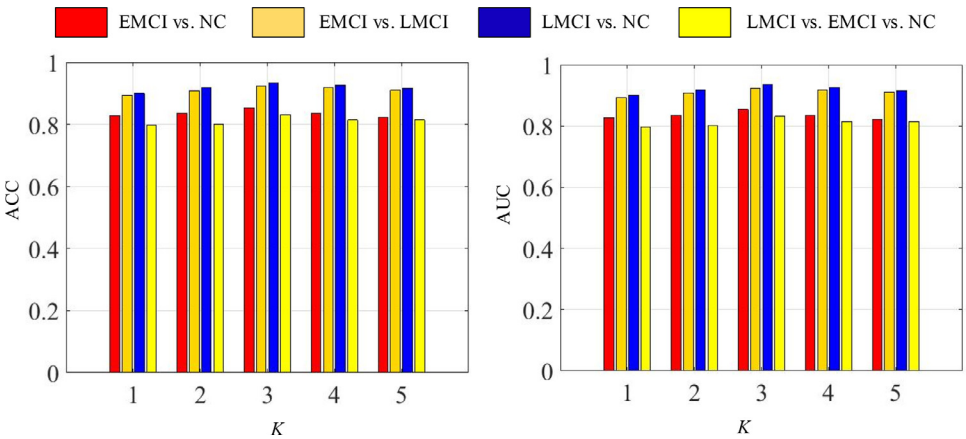

Fig. 6. Mean detection accuracy and AUC using different K orders in tasks of EMCI vs. NC, EMCI vs LMCI., LMCI vs. NC, and LMCI vs. EMCI vs. NC.

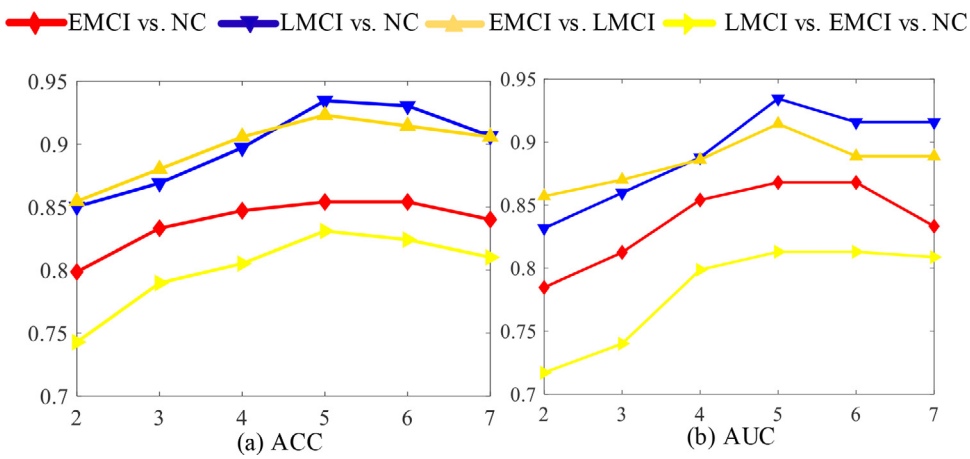

Fig. 7. The detection performance of our model with different step of random walk in different tasks.

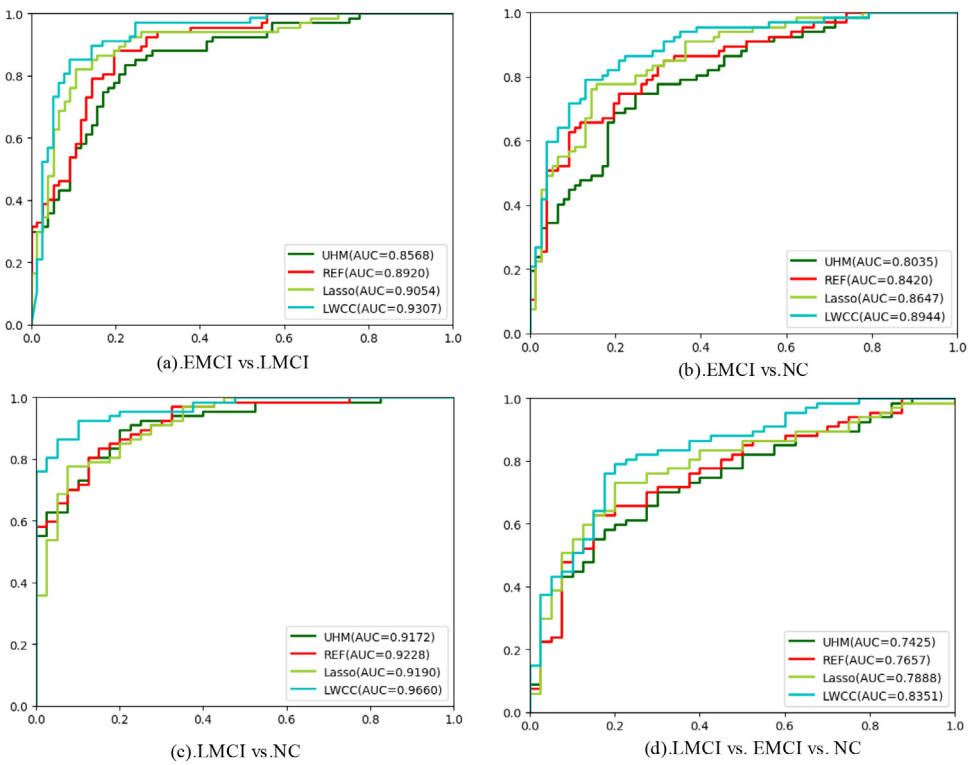

Fig. 8. The performance comparison of using different feature extraction in each detection task.

model at different scales and select the best scale p from the average cross-validation results. Fig. 7 shows the performance of our model using different steps of random walk in different tasks. The average classification performances at different scales of p are as follows: when $p = 2$, the ACC performance of EMCI vs. NC, LMCI vs.NC, EMCI vs. LMCI and LMCI vs. EMCI vs. NC are 79.86%, 85.04%, 85.47% and 74.28%, respectively. When $p = 3$, the corresponding results are 83.33%, 86.91%, 88.03% and 78.98%. When $p = 4$, the corresponding results are 84.72%, 89.72%, 90.59% and 80.52%. When $p = 5$, the corresponding results are 85.42%, 93.46%, 92.31% and 83.11%. When $p = 6$, the corresponding results are 85.42%, 93.05%, 91.45% and 82.42%. When $p = 7$, the corresponding results are 84.02%, 90.65%, 90.59% and 81.02%. Therefore, we set the optimal p as 5. We note that utilizing random walks by setting $p > 1$ improves model performance due to utilizing additional information. However, the improvement of model accuracy is not endless. When p gradually increases (e.g., p>4), the number of layers and parameters of GCN also increase, and the performance of the model will be suppressed to some extent.

5.4. The influence of LWCC

Extracting effective features is a key to detect MCI. In this study, we use functional and structural information in connectivity networks integrated via LWCC, which are concatenated as the feature vectors to represent the vertices of a population graph. Compared with traditional methods, our method not only reduces the feature dimension, but also quantifies the clustering of each node in the weighted network. To demonstrate the effectiveness of LWCC, we compare it with the same model using other feature selection methods. Specifically, we compare LWCC with: (1) Extract the upper half matrix (UHM) of the brain network [32]; (2) Recursive feature elimination (RFE) [53]; (3) LASSO [54]. Fig. 8 shows the performance of the proposed model using different feature extraction

Table 4
Mean detection performance of different methods in the three tasks (%).

EMCI vs.NC					LMCI vs.NC				
Method	ACC	SEN	SPEC	F1	Method	ACC	SEN	SPEC	F1
UHM	80.25	78.62	81.58	82.65	UHM	88.58	83.67	91.59	92.33
REF	83.68	82.51	84.16	86.88	REF	91.25	90.65	92.38	95.58
Lasso	83.96	86.25	80.21	85.50	Lasso	90.68	91.29	90.35	94.67
LWCC	85.42	86.57	84.42	89.98	*LWCC*	93.46	94.03	92.50	96.87
EMCI vs. LMCI					EMCI vs. LMCI vs.NC				
Method	ACC	SEN	SPEC	F1	Method	ACC	Precision	Recall	F1
UHM	85.29	83.42	86.34	87.47	UHM	73.16	70.67	74.58	75.87
REF	88.48	89.28	87.65	90.17	REF	80.28	78.25	82.65	79.28
Lasso	89.65	91.58	86.79	91.65	Lasso	79.50	80.79	79.24	81.85
LWCC	92.31	93.51	90.00	93.25	*LWCC*	83.11	81.30	85.65	83.62

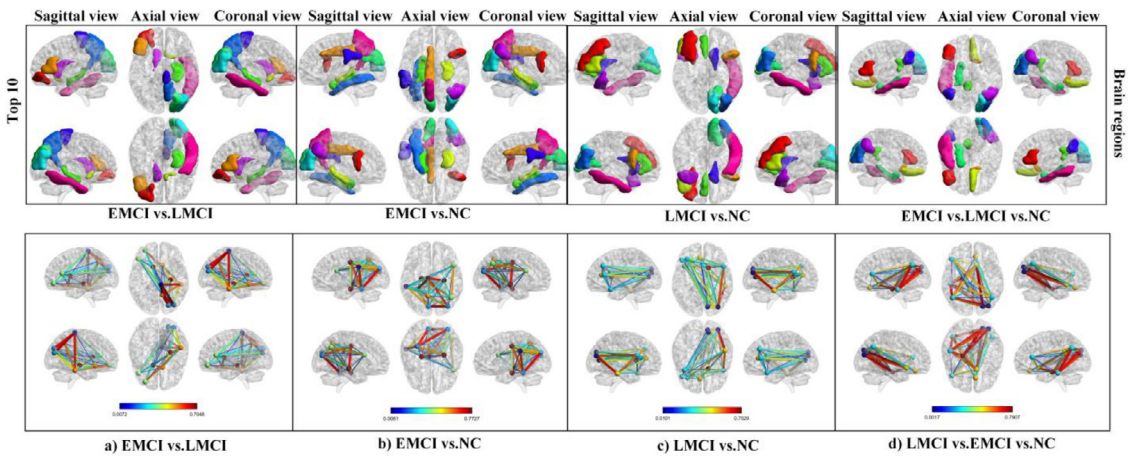

Fig. 9. Top 10 brain regions and functional connectivity of the four tasks using our proposed method.

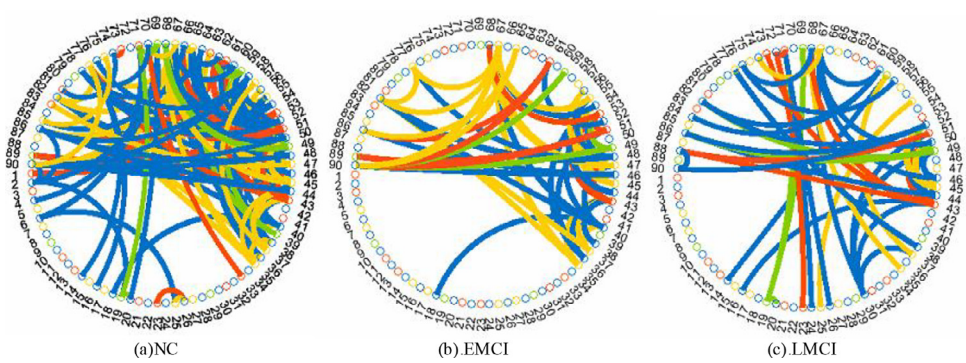

Fig. 10. The structural connectivity of our method.

methods in different detection tasks. More detailed numeric results are shown in Table 4, which also demonstrates the effectiveness of extracting network features using LWCC.

5.5. Most discriminative regions

Different parts of the brain are responsible for different tasks and not all regions are closely related to MCI [55]. Hence, we attempt to utilize our proposed method to search for these relevant ROIs for a better understanding of brain abnormalities. It is meaningful for us to find useful information from the biomarkers to discover the underlying relationships for disease detection. We shield one brain regions in turn and then get 90 detection results. According to the corresponding detection performance, we analyze the impact of the corresponding ROI to the disease. Fig. 9 shows the top 10 brain regions and functional connectivity of different tasks using our proposed method. The results show that inferior frontal gyrus, olfactory cortex, parahippocampal gyrus, middle occipital gyrus, and inferior temporal gyrus are highly correlated with MCI, because these regions are found in all four classification tasks. In addition, we visualize the connections between brain regions in patients at different stages. Fig. 10 shows the structural connectivity of our method, which indicates that the structural connections of brain regions are different in patients at different stages. Fig. 11 shows the examples of the brain changes during the time and the influence of different stages of the disease.

5.6. Algorithm comparison with related works

Apart from investigating parameters of proposed model impact prediction performance, we further compare our MSE-GCN method with other different competing methods in related work.

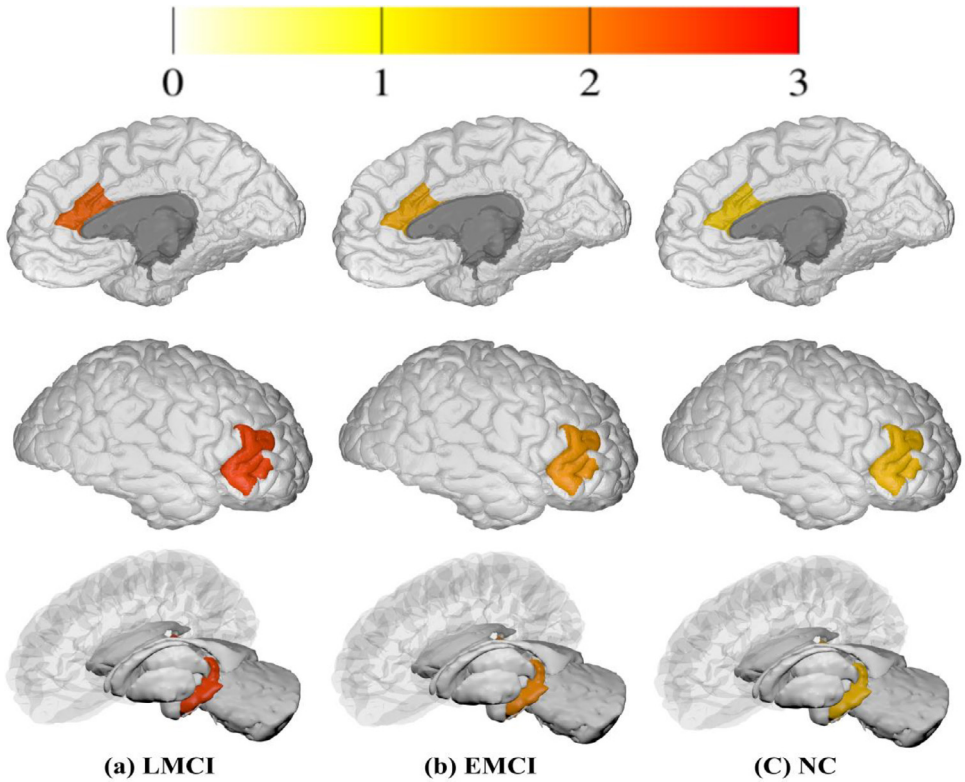

Fig. 11. The examples of the brain changes during the time and the influence of different stages of the disease.

Table 5
Performance comparison with literature studies.

Method	Modality	Subject	ACC	SEN	SPEC	AUC
Zhao et al. [43]	R-fMRI+DTI	77 EMCI, 40 LMCI,67 NC	78.4	84.7	N/A	0.81
Xing et al. [33]	R-fMRI+MRI	191 EMCI, 177 NC	79.73	86.49	72.97	N/A
Wee et al. [56]	fMRI	29 EMCI, 30 NC	79.6	75.8	70.0	N/A
Yu et al. [57]	fMRI	50 MCI, 49 NC	84.8	91.2	78.5	N/A
Zhu et al. [2]	MRI + PET + CSF	99 MCI, 53NC	83.5	95.0	62.8	N/A
Proposed	R-fMRI+DTI	77 EMCI, 40 LMCI, 67NC	85.42	86.57	84.42	86.47

Table 5 shows the algorithm comparisons with the related algorithms. We can see that our proposed method has achieved overall better performance in MCI detection. Apart from better prediction performance, our proposed method does not need to construct a complex brain connection network, that may be better adopted in other prediction tasks.

6. Conclusions

In this study, we propose a novel MCI detection framework based on MSE-GCN. Specifically, we extract the LWCC from FC and SC, and fuse the LWCC of FC and SC to construct a sparse graph viaintegration of the age and gender information. Then,we use an enhanced MS-GCN model with random walk embedding to detect MCI. Four groups of detection experiments and comparison with the state-of-the-art methods have been conducted on ADNI dataset, the results show that the enhanced MS-GCN model with random walk embedding can increase receptive field of GCN and get significant node representations, which is beneficial to improve detection performance, thus the proposed method can be an effective reference for early clinical diagnosis of MCI. However, the current study still has some limitations. First, our method relies on the subject's age and gender, and we have not considered other biological indicators or clinical cognitive neurology scales. Second, our model does not consider the longitudinal time point data, while the longitudinal related research has shown the ability to reveal disease progression and has significant research value [58]. The purpose of this study is to achieve early detection/classification of AD. However, the clinicians may further need to classify a variety of different brain diseases. The proposed method can be extended to other brain disease classification and it is worth further exploration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

ADNI dataset (http://adni.loni.ucla.edu).

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